

GLIDE: Guided Learning in Immersive Digital Environments

International Space Station Experience App

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1 Introduction

In many scientific fields, students can directly interact with each environment. Biology is right out the door, computer science sits on our devices, physics is literally everywhere, and psychology is in the people around us. There is one field that struggles to maintain that level of direct interaction due to its inaccessibility. That space-related field. Most space-related scientific fields are entirely inaccessible to students in most environments [1]. On top of that problem, traditional teaching methods about these environments often rely on static tools that limit user engagement, such as textbooks, PowerPoint presentations, and videos [2]. However, both of these problems, the inaccessibility and poor teaching, can be solved with the advancement of virtual reality (VR) technologies. VR environments allow students to have more opportunities to engage in immersive interactive experiences that work with, not against, their interests [3]. Specifically in the space-related field, VR technology can bring these fields, almost in their full capacity, right in front of their eyes. Our specific environment, focused on the International Space Station (ISS), can help foster engagement and interest in a field that is difficult to interact with daily.

To bridge this gap, this project develops a VR simulation of the ISS designed for accessibility and to develop a deeper understanding and interest in the STEM fields in space. Users can explore the station in a 3D environment, interacting with key components through intuitive ways such as raycast-based object selection and integrated text-to-speech (TTS) descriptions, or experience a guided tour that ensures the user can understand the entirety of the space station. These features shift the typical learning experience from a passive observer to an active, hands-on learning experience, allowing users to build intuitive knowledge of the space station that aligns with their interests [3].

2 Related Work

2.1 Development of Interactive Media on Science for Junior High Students

This study focused on the development and validation of interactive learning media for biotechnology education in Papua, Indonesia, a region where limited facilities and unconventional learning methods is detrimental to educational quality. The research done demonstrated that students using interactive digital media showed significant improvements in learning outcomes, with some groups seeing a 15-point increase in comprehension compared to a 4-point increase in traditional settings [9]. These findings suggest that inter-

active tools can effectively bridge educational gaps by increasing student motivation and self-efficacy, even in resource-constrained environments.

Beyond specific regional studies, the design of interactive science media for younger audiences has shifted toward sandbox exploration, which helps kids to explore things they're curious about. Modern frameworks suggest that for students to truly grasp abstract concepts, apps should prioritize visual-spatial interaction, such as 3D model rotation and active discovery, over static text and passive video viewing [10]. By allowing students to interact with virtual variables in a stress-free environment, developers can develop a scientific mindset where experimentation is prioritized over memorization. This transition from passive consumption to active manipulation ensures that digital media acts as an entry point for deeper scientific engagement.

2.2 Immersive Learning and Voice Interaction

Virtual Reality (VR), such as the International Space Station (ISS) environment used in this project, offers a unique way to learn by doing. Being "inside" a simulation helps students connect with the material in a way that reading a textbook cannot. Research shows that these immersive spaces make students feel more like designers and engineers, which increases their emotional stake in solving the problems presented to them [7].

However, VR can be overwhelming if the screen is cluttered with menus and text. Using a voice-based assistant solves this by letting the student keep their hands and eyes on the task. Research indicates that successful VR education depends on a safe, distraction-free space where students can experiment [8]. By using voice interaction, we can provide helpful tips in small, manageable pieces, keeping the focus on the environment while the assistant serves as a "guide on the shoulder."

2.3 Interactive Digital Modules for Space Education

The transition from static lectures to digital space education has been extremely effective for younger learners who find space concepts confusing or difficult to visualize. A 2025 study on primary school students utilized a digital space module featuring nine different educational games to teach the structure and appearance of the Earth and the history of space exploration [11]. The results showed that interactive multimedia, when equipped with hands-on activities, greatly increased student knowledge and long-term interest in space-related careers compared to traditional learning methods.

In addition to basic knowledge acquisition, modern space-themed applications are increasingly focused on "Playlearning" frameworks that build foundational STEM skills through exploration. For example, recent collaborations between NASA and educational developers have demonstrated that structured, self-paced learning, where children explore the solar system through interactive simulations, can lead to a 90% improvement in core science skills for early learners [12]. By allowing students to build their own space missions or rotate photorealistic 3D models of spacecraft, these tools allow ease of entry for complex science, fostering a sense of agency that makes students feel more like active scientists rather than passive observers.

3 Methodology

This project was completed in four phases, as seen in Figure 1 below, and further explained in the following sections. The team collaborated using a Trello workspace to track progress, post helpful ideas, and set completion dates for each of the phases.

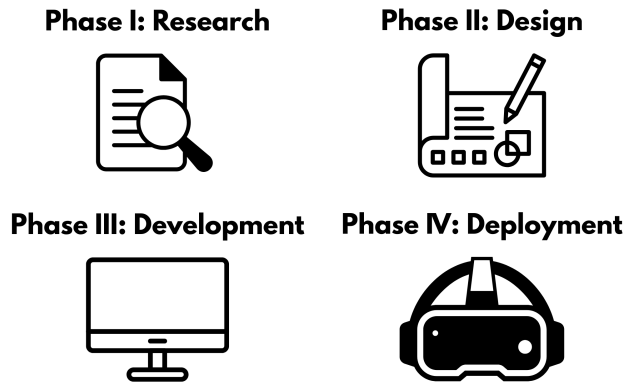


Figure 1: The Four Phases

3.1 Research Phase

The research phase consisted of two separate research tasks taken on by the team. Task 1 was to study and gain information about the International Space Station itself. Using several online resources, we were able to develop concise yet informative paragraphs about the different modules and objects we showcase within the digital environment. Task 2 was to learn how to use the Unity Game Engine to develop VR applications. This included learning the C# coding language, building models in VR, and deploying applications to VR headsets. Another important piece in task 2 was learning how to integrate AI tools with both Unity and the headset to enhance the overall app experience.

3.2 Design Phase

After a couple weeks of research and preparation, the team entered the Design phase. In this phase, we determine the scope of our project and what we hoped to achieve in the time allotted to us. We assigned different levels of priority to different aspects of the project, as seen in Table 1. Tier I of the project consisted of the most essential features we hoped to include, such as building the actual environment for the app, realistic movement physics, interactable objects, and Text to Speech (TTS) capabilities. Tier II features were ones we hoped to and planned on including from the start but would require a lot of time to implement, and thus were not considered essential elements. The Guided Tour Mode, for example, is a great feature, but not as important as a working version of all of the Tier I tasks. Tier III features are mostly quality-of-life improvements to enhance the experience overall, but do not contribute much to the content of the application.

3.3 Development Phase

The Development Phase made up the majority of the time spent working on this project. We started from a model of the International Space Station (ISS) provided by NASA on their GitHub page. From there, we applied the given textures and added collision to all the objects in the space station, as well as an outer sky-box to be viewed from windows in the station. Then we moved on to player movement, creating a player object and a movement script meant to mimic zero gravity like on the real ISS. The next priority was to add interactive elements to the design, which involved adding ray-interactable scripts to the player and objects to allow users to select and learn more about the space station. The final Tier I task was to enable TTS for both

Priority	Task
Tier I	• Virtual Environment with Full Textures • Realistic Movement Features • Interactable Objects • TTS Capabilities for Room and Object Descriptions
Tier II	• Implementing Guided Tour Mode • Adding Player Hands • AI Assistant for Further Learning (Incomplete)
Tier III	• Title Screen, Pause Menu, and End Scenes • Ambient Sounds

Table 1: Visual demonstration of the Priority Levels assigned to different parts of the project

the interactable objects and when entering a new room. We utilized Wit.AI to handle the actual conversion of written messages to audio, but constructed our own audio system with room and object triggers.

Once this basic version was complete, we moved on to our Tier II tasks. The first was an update to the player object to add in virtual hands to help users better interact with the objects in the space station. But the main Tier II priority was the Guided Tour Mode. While all the previously worked on features exist in both the Free Roam and Guided Tour modes, the latter expands the learning capabilities of our application. The guided tour includes features such as arrows to point users between rooms, locking and unlocking doors to restrict access to one room at a time, and preventing other audio from playing over room narrations. The last Tier II task was to build in an AI Assistant to allow users to ask further questions about what they are experiencing to enhance their learning further than what pre-made scripts can provide. Unfortunately, due to time constraints the AI assistance features were not able to be fully implemented.

The final quality-of-life tasks from Tier III we added were a title screen environment to choose between modes, an end scene for the Guided Tour, and a pause menu to allow users to change the volume of the audio and quit the app. We also added ambient noise that would be common on the space station, as well as reactive audio when a player collides with any object.

3.4 Deployment Phase

Throughout the entire process, there was constant testing and version control using a shared GitHub repository, but the final version of the app was tested for bugs and errors using the remaining time on the project. While some issues may persist, the essential aspects of the application and the learning capabilities should not be affected.

4 Tools and Technologies

There were many different tools and technologies used in the methodology of our project that are listed and explained in this section.

4.1 Unity Game Engine

The project was built on Unity version 6000.3.8f1. Outside of the built-in Unity Packages, we used the Meta XR Platform, the Skyboxes - Space assets and the Low Poly 3D Icons assets packages. Since the project was made for VR, we used the Android build platform.

4.2 Meta Quest 3 Headset

We used a Meta Quest 3 VR headset for testing and deploying the final project on for demonstration.

4.3 Wit.AI

Wit.AI is a service through Meta to handle TTS messages, which made it easy to integrate with our Unity project and the Meta Quest 3 headset.

4.4 VS Code

The team decided to use the VS Code software for all coding-related tasks due to its easy integration with both Unity and GitHub, and our prior familiarity with it.

4.5 ChatGPT

We used ChatGPT mainly to help with error debugging.

4.6 GitHub

The team used a shared GitHub repository to ensure version control as we worked on different parts of the project simultaneously. This is now available for public viewing for those interested in viewing the project in its current complete state or expand the features and improve the application beyond what we finished in the allotted time we had.

4.7 Trello

The team collaborated using a Trello Workspace to create weekly goals, assign tasks to team members, share helpful links and resources, and track overall progress.

5 Results

The completed Virtual Reality (VR) Space Station system provides an immersive and interactive learning environment, allowing users to explore and engage in a fully developed 3D space station. The space station has the option to have learning and free roam, which allows users to move around on their own and be given helpful instructions in each room.

Table 2 summarizes the main functions of the VR application.

When the user launches the space station, a title screen room appears, which lets users choose between a learning mode and a free-roaming mode. In the free-roaming mode, users can explore the station at their own pace. In the learning mode, the system is being developed to provide information when users enter different rooms.

After an option is selected, users are transported into the space station environment, which has been made to move around smoothly. Users are able to move freely throughout the space station while staying

Environment/Screen	Function
Title Screen Room	Allows users to select between learning mode and free-roam mode
Space Station Interior	Fully explorable VR environment with multiple rooms
Guided Tour Mode	Provides guided, room-based educational content spoken aloud
Free-Roam Mode	Enables unrestricted exploration of the station
Interactive Objects	Allows users to select and examine items for more information

Table 2: Overview of System Features and Interfaces



Figure 2: Title Screen Room for Mode Selection

within the set boundaries we have created. The interior layout of the station is complete, allowing users to navigate through different rooms in a 3D space.

The completed layout of the interior of the space station allows users to walk through various sections of the station, each modeled to resemble a realistic space station environment. The design has both visual engagement and functional learning spaces. Within the environment, Learning Mode enhances user engagement by providing contextual information based on location. As users enter different rooms, relevant educational content is presented, allowing them to understand the purpose and function of each area within the station.

In addition to guided learning, interactive objects are integrated throughout the station. Users can select and examine these objects to receive additional information, reinforcing learning through interaction.

The system was designed to include a fully implemented large language model (LLM) assistant, enabling users to ask questions and receive detailed explanations through continuous conversation. This assistant was intended to allow users to explore objects and concepts within the environment in a more interactive and guided manner, activated by speaking the wake word “assistant” within a sentence. However, due to unresolved microphone detection issues on the Oculus device, the voice input functionality did not operate reliably during testing. Additionally, time constraints during the project period prevented further debugging and resolution of this issue. As a result, the assistant feature could not be effectively utilized in practice, limiting user interaction with the system.

Overall, the final system demonstrates a successful integration of immersive VR exploration with in-

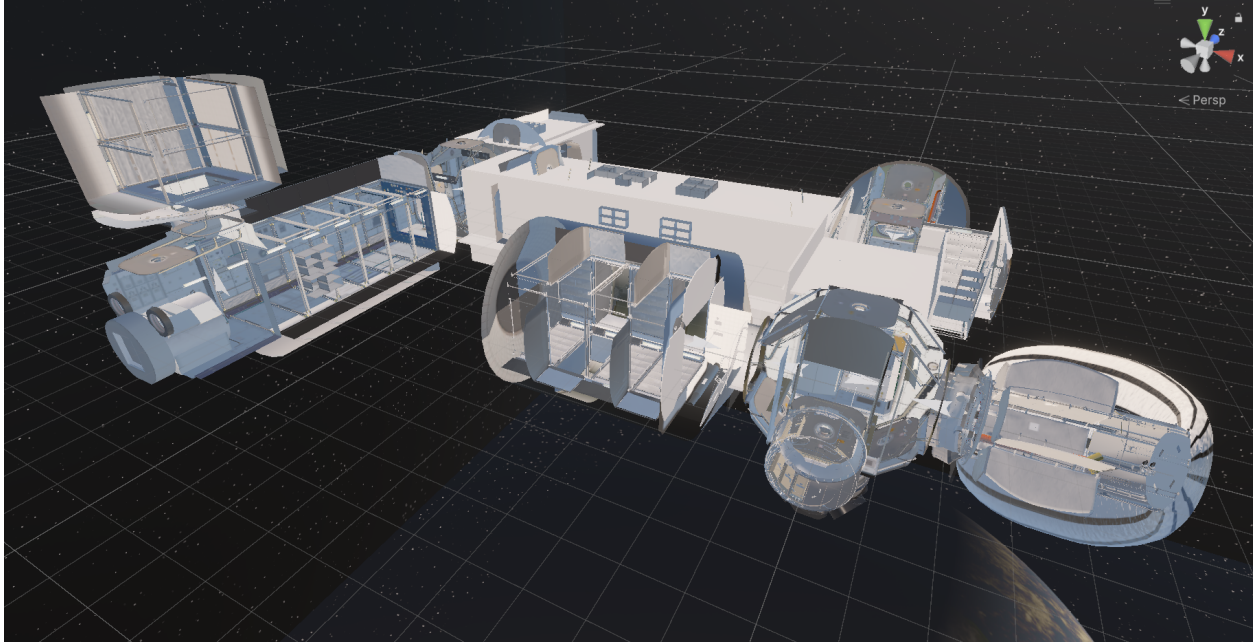


Figure 3: Exterior Overview of the VR Space Station



Figure 4: Interior View of Space Station

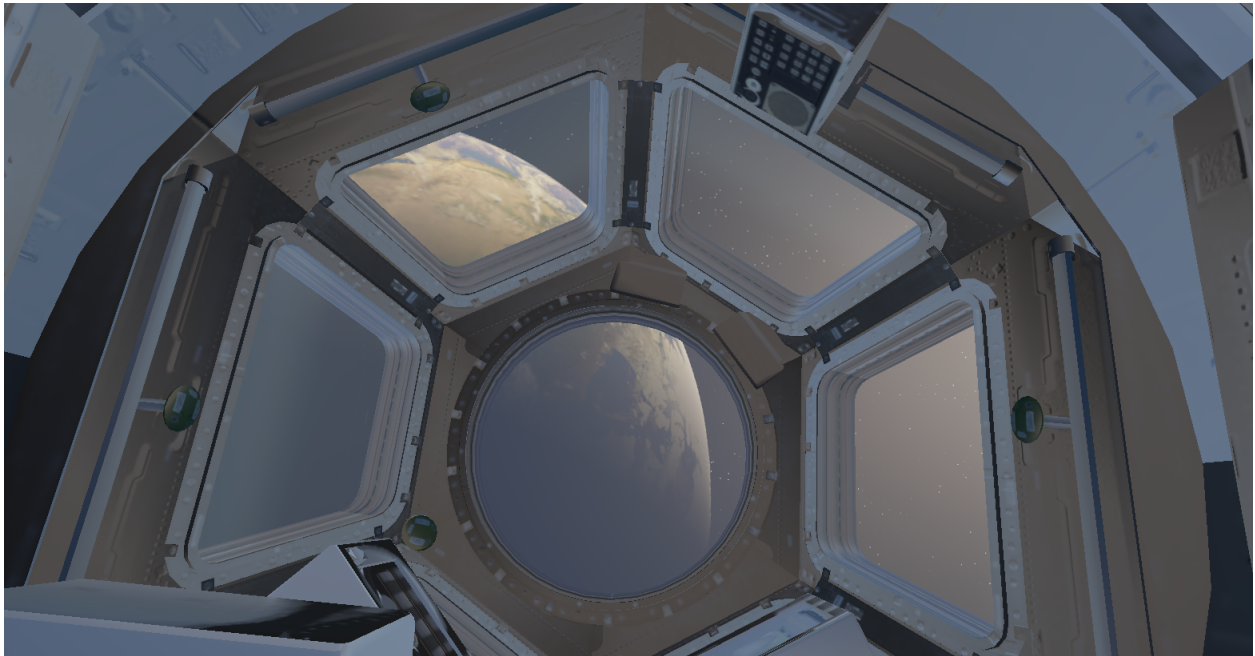


Figure 5: Interior View of Space Station

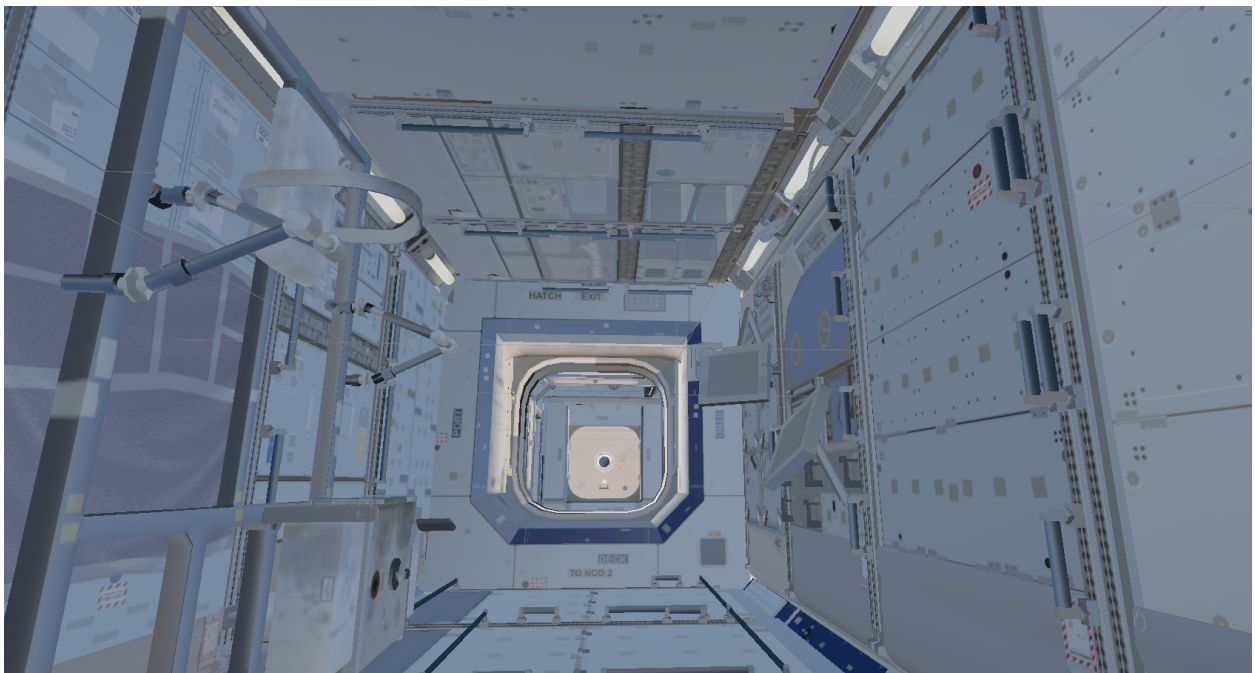


Figure 6: Interior View of Space Station

teractive features. By combining navigation, object interaction, and engaging narration, the application provides an educational experience that supports both independent exploration and helpful instruction.

6 Conclusion

In conclusion, this project demonstrates the effectiveness of virtual reality as a tool for enhancing engagement and accessibility in STEM education. By creating an interactive VR simulation of the International Space Station, the project overcomes the problems of traditional static learning tools and gives users an immersive, self-paced way to explore a complex scientific environment that is otherwise unreachable.

Including features such as raycast-based object selection and text-to-speech guidance changes the experience from passive observation to active, hands-on participation. This not only increases the users' understanding of the ISS and its operations but also sparks curiosity and increases self-led learning.

While the current prototype is a good start, future improvements could include adding interactive ISS systems (dynamic simulations of life support or robotics), incorporating AI-driven guidance, and conducting user evaluations and case studies.

Ultimately, this project demonstrates the potential of virtual reality technologies in immersive learning. As VR continues to develop, simulations hold promise for increasing access to advanced scientific environments and inspiring the next generation of space explorers and STEM researchers.

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